

Week 12 Tutorial – Time Series - MEMO

QUESTION 1

- a) Yes. There is no long-term trend (i.e. the mean is constant), and the variance appears relatively constant throughout the series. It is thus plausible that this time series is stationary.
- b) Yes. The correlogram (i.e. sample ACF) clearly reveals autocorrelation values at lags 1, 2, 3 and 4 that are significantly different from 0. Hence, there is autocorrelation present in the time series.
- c) The original time series. Since the original time series appears stationary, and the correlogram confirms that it is plausible that it is stationary, there is no need to transform the series.
- d) Since we haven't transformed the series by applying differencing, we will apply a model directly to the original data. The behaviour of the sample ACF and PACF plots suggests that an AR(1) model would be appropriate to fit to the data as the correlogram shows slow decay to 0 as the value of the lag increases, and the PACF has one significant spike at lag 1 and then abruptly shuts off.
- e) A non-seasonal first-order autoregressive, AR(1), model.

(NB NOTE: an AR(1) model can be equivalently expressed as an ARIMA(1,0,0) model – they are all the SAME thing!)

- f) $X_t = \phi_0 + 0.6244X_{t-1} + Z_t$
- g) $\mu = \frac{\phi_0}{1-\phi_1} \Rightarrow \phi_0 = \mu(1-\phi_1) = 80.0476(1-0.6244) = 30.0659$
- h) $\sigma^2 = 0.9136$
- i) For an AR(1) model, $V[X_t] = \frac{\sigma^2}{1-\phi_1^2} = \frac{0.9136}{1-(0.6244)^2} = 1.497$

$$\rho(1) = \phi_1 = 0.6244$$

j) Residual diagnostics are carried out in order to determine whether the random noise/error series (Z_t) can plausibly follow a white noise process. If they can, then the model is deemed to be an adequate fit to the data as no autocorrelation in the residuals means that the model has captured all of the relevant information/patterns in the data. If the residual diagnostics provide evidence that the error series does not follow a white noise process, then the model is not an adequate fit to the data as it has not captured all of the relevant information/patterns in the data.

- k) The non-seasonal ARIMA(1,0,0) (i.e. AR(1)) model is an adequate fit to the data, as the residual diagnostics provide evidence that the error series could follow a white noise process:
- The plot of the standardised residuals does not show any pattern, it has a relatively constant variance and a mean of 0
 - The sample ACF of the residuals does not show any significant spikes at any lags, implying that there is no autocorrelation in the residual series
 - The Normal Q-Q plot does not demonstrate any significant deviation from the straight line, indicating that it is plausible that the residuals are normally distributed
 - The p-values of the Ljung-Box tests for autocorrelation (at each lag up to lag 20) are all > 0.05 thus providing more evidence that there is no autocorrelation in the residual series

Thus we can conclude that the residuals are white noise, and that the model has captured all the relevant information/patterns in the data.

QUESTION 2

a) The time series plot can be examined to reveal:

- What components are present in the time series that give rise to the pattern in the data
- The behaviour of the mean and variance of the time series
- Whether or not the time series is stationary
- Outliers

b) The following should be taken into account in determining the orders of p , d , and q :

- The first thing to take note of is whether the time series is stationary or not. If it is not (i.e. it contains a trend and/or changing variance) then it must be transformed. If you have to apply differencing to remove the trend or seasonality, then take note of what order of differencing is applied. (Recall that in this course we only deal with linear trends and/or linear trends with seasonality and thus only apply first-order differencing. So, $d = 1$, if we do difference the series). If we do not need to difference the series, then $d = 0$.
- Once the transformed time series appears stationary, we would plot the correlogram and partial correlogram and study their behaviour to see if there is any other information (other than the removed trend/seasonality) that we can capture from the data:
 - If the correlogram shows a sharp cut off after lag q , and the partial correlogram shows a slow decay to 0, then an $MA(q)$ may be appropriate to apply to the data. Hence, $p = 0$ and $q = q$.
 - If the correlogram shows a slow decay to 0 and the partial correlogram shows a sharp cut off after lag p , then an $AR(p)$ model may be appropriate to fit to the data. Hence, $p = p$ and $q = 0$.
 - If both plots show slow decay to 0, then we would fit a few low-order $ARIMA(p,d,q)$ models to the data e.g. $ARIMA(1,1)$, $ARIMA(2,1)$, $ARIMA(1,2)$, $ARIMA(2,2)$. We would then compare these models using the AIC and BIC and select the model with the smallest values across the AIC and BIC. Hence, $p = p$ and $q = q$ of the model that is chosen.

c) You should look at:

- Plot of standardised residuals
- Normal Q-Q plot of standardised residuals
- ACF plot of residuals
- p-value(s) of Ljung-Box test(s)

You examine all the above in order to make a decision as to whether it is plausible that the residuals (and hence the error series that they represent) follow a white noise series. If it is plausible, then you can conclude that the $ARIMA(p,d,q)$ model that you have fitted has captured all the relevant information from the data (and is thus an adequate fit to the data). If it is not plausible, then the fitted model has not captured all the relevant information/patterns in the data (is not an adequate fit to the data).

QUESTION 3

- a) Yes. There is no long-term trend or seasonality so the mean is independent of time/constant/not changing with time, and the variance appears relatively constant (no large changes over time or gradual increase or decrease over time).
- b) Both the sample ACF and PACF show slow decay to 0 as the lag increases. Considering there is no seasonality present in the data, this behaviour is indicative of a non-seasonal ARIMA(p,d,q) process.
- c) The orders p and q are not apparent from the behaviour of the plots, so we would fit a few low-order non-seasonal ARIMA(p,d,q) models.
- d) Information criteria essentially trade-off the goodness-of-fit of a model with the order (number of terms and corresponding parameters i.e. complexity) of the model. The smaller the value of the AIC, the better the fit of the model **compared to the other models fitted to the same data**.
- e) The ARIMA(2,0,1) model has the smallest AIC value and is thus the best model out of the four to proceed to fit to the data.
- f) The next step would be to carry out the diagnostic checks of the residuals of the fitted ARIMA(2,0,1) model.

This would be done by examining the plots of the standardised residuals, the normal Q-Q plot/histogram of the standardised residuals, the ACF of the residuals and the plot/p-values of LjungBox tests up to a certain lag.

If all of the above plots provide evidence that the residuals follow a white noise process, then you would conclude that the ARIMA(2,0,1) model has captured all the relevant information in the data and proceed to use it to forecast future values of the time series.

If even one of the above plots/diagnostics indicated that one of the assumptions of a white noise process had been violated, you would conclude that the ARIMA(2,0,1) is not an adequate fit for the data, and you would return to step 4 of the modelling process to suggest a different model/s or try and improve the current fitted model.

QUESTION 4

A

- i) The sample ACF cuts off after lag 4.
- ii) The sample PACF plot cuts off after lag 2
- iii) We would fit a few low-order ARIMA models here e.g. ARIMA(1,1), ARIMA(2,1), ARIMA(1,2), ARIMA(2,2)
- iv) We would fit a few low-order ARIMA models here e.g. ARIMA(1,1, 1), ARIMA(2,1,1), ARIMA(1, 1, 2), ARIMA(2, 1, 2)

B

- i) The sample ACF decays slowly to 0 as the lag increases
- ii) The sample PACF plot cuts off abruptly after lag 1
- iii) An AR(1) model would be appropriate here
- iv) An ARIMA(1,1,0) model would be appropriate here

C

- i) The sample ACF cuts off abruptly after lag 1
- ii) The sample PACF plot decays slowly to 0
- iii) A MA(1) model would be appropriate here
- iv) An ARIMA(0,1,1) model would be appropriate here

D

- i) The sample ACF cuts off after lag 2
- ii) The sample PACF plot decays slowly to 0
- iii) A MA(2) model would be appropriate here
- iv) An ARIMA(0,1,2) model would be appropriate here

E

- i) The sample ACF decays slowly to 0
- ii) The sample PACF plot decays slowly to 0
- iii) We would fit a few low-order ARIMA models here e.g. ARIMA(1,1), ARIMA(2,1), ARIMA(1,2), ARIMA(2,2)
- iv) We would fit a few low-order ARIMA models here e.g. ARIMA(1,1, 1), ARIMA(2,1,1), ARIMA(1, 1, 2), ARIMA(2, 1, 2)

F

- i) The sample ACF decays slowly to 0 as the lag increases
- ii) The sample PACF plot cuts off abruptly after lag 3 (the significant spike at lag 11 may just be significant due to chance)
- iii) An AR(3) model would be appropriate here
- iv) An ARIMA(3,1,0) model would be appropriate here

QUESTION 5

A

- i) This is an AR(1) process
- ii) Since there is no constant term, ϕ_0 , we can assume $\phi_0 = 0$ which then implies that the mean is 0 as for an AR(1) stochastic process $\mu = \frac{\phi_0}{1-\phi_1} = \frac{0}{1-0.62} = 0$
- iii) $\rho(1) = \frac{\phi_1 V[X_t]}{V[X_t]} = \phi_1 = 0.62$
- iv) $V[X_t] = \frac{\sigma^2}{1-\phi_1^2} = \frac{(2.21)}{1-(0.62)^2} = 3.590$

B

- i) This is a MA(1) process
- ii) $\mu = 22.45$
- iii) $\rho(1) = \frac{\theta_1}{1+\theta_1^2} = \frac{-0.49}{1+(-0.49)^2} = \frac{-0.49}{1.2401} = -0.395$
- iv) $V[X_t] = \sigma^2(1 + \theta_1^2) = (1.01)(1 + (-0.49)^2) = 1.253$

QUESTION 6

- a) Increasing, non-linear trend; random variation and seasonality.
- b) It is non-stationary because its mean is not constant due to the presence of the upward trend and the seasonality. The variation in the data appears to be relatively constant, so that is not a concern with respect to the stationarity of the time series.
- c) This is a common practice in all predictive (i.e. forecasting) statistical analyses. The idea behind it is that we need to “keep back” a small proportion of the data (the test data) so that we can test the accuracy of the model that we fit on the rest (the majority – approx. 80%) of the data which is called the training data. If we did not do this then we could not know how well our model/s perform when forecasting into the ‘future’.

QUESTION 7

a)

A: plausibly stationary (no trend and variance and variance looks relatively constant)

B: Non---stationary (clear upward linear trend)

C: plausibly stationary (no trend and variance looks relatively constant)

b)

A: sample ACF shows slow decay to 0, sample PACF has one significant spike at lag 1, and then abruptly cuts off. This implies that an $ARIMA(1,1,0)$ i.e. an $AR(1)$ model would be appropriate to fit to the data

B: The time series appears relatively constant in variance, so a variance stabilising transformation is not necessary. The trend is clearly linear, so differencing of order 1 at lag 1 would be an appropriate transformation to remove the trend.

C: The correlogram appears to cut off abruptly at lag 1, but there is also a non-zero autocorrelation at lag 18 and lag 19. The PACF slowly decays to 0. So one would fit a few low-order $ARIMA$ models (one of which would be an $ARIMA(0,1,1)$ i.e. a $MA(1)$ model).

QUESTION 8

Mean: $E[X_t] = \mu = 8.372$

Variance: $V[X_t] = \sigma^2(1 + \theta_1^2) = 5.312(1 + 0.8^2) = 8.712$

Autocorrelation at lag 1: $\hat{\rho}(1) = \frac{\theta_1}{1+\theta_1^2} = \frac{0.8}{1+0.8^2} = 0.488$